Design Rationale --- Esong AI Video Platform

Overall approach

The three prototypes deliberately represent **three different relationships between a creator and AI**, rather than three visual skins of the same editor.

- **Prototype 1 --- Code Flow:** AI removes complexity for a casual creator.
- **Prototype 2 --- Code Flow Pro:** AI accelerates an experienced creator while keeping the human in control.
- **Prototype 3 --- Code Flow Signal:** AI helps a creative explorer discover possibilities before deciding what video to make.

This differentiation follows the assignment's target-user, device-priority, AI-interaction, visual-complexity, and navigation axes. All three remain frontend prototypes: AI generation, video composition, and export are simulated with mock data rather than real backend/ML processing.

Prototype 1 --- Code Flow

Target: Casual creator, with a mobile-first experience.

Color: light + colorful scheme

The central design principle is **"AI removes complexity rather than exposes it."** The user may begin with a template or a short description of an idea, then move through a guided flow: **Idea** → **Storyboard** → **Scenes** → **Character** → **Video**.

The interface uses a minimalist/friendly visual language, card-based content, progressive disclosure, and approachable language. A hybrid wizard/template model avoids the two extremes of a completely blank prompt and a completely restrictive template system. Suggestion chips and simple controls reduce the cognitive load of deciding what to do next.

The storyboard uses a card-and-overview model: cards make individual scenes easy to browse and edit, while the overview makes sequence and

duration understandable. Scene editing is exposed through a focused bottom-sheet style interaction rather than a dense professional inspector. Reordering supports both touch-oriented drag interaction and explicit movement controls.

Trade-off: Code Flow intentionally hides professional video concepts and exposes only the controls needed at each stage. This makes the experience approachable and mobile-friendly, but it is less efficient for an expert who wants simultaneous access to many production parameters. That limitation is intentional and creates a clear contrast with Prototype 2.

Prototype 2 --- Code Flow Pro

Target: Professional creators, filmmakers, editors, and marketing-oriented content creators; desktop-first.

Color: dark theme (professional oriented)

The core principle is **"AI accelerates creative decisions without taking creative control away from the creator."** Unlike Prototype 1, the workspace deliberately exposes useful professional concepts such as scenes, camera, lighting, timeline, audio, variations, and creative direction.

The layout follows a production-workspace mental model: project navigation, a large video preview, a persistent Inspector on large screens, a multi-track timeline, and contextual AI actions. The Inspector keeps the selected scene's properties visible, while the timeline connects scene-level changes to the overall sequence.

AI is designed as a **copilot rather than an autonomous editor**. Variations are non-destructive: the creator can preview a generated alternative and explicitly choose whether to use it. Copilot changes are presented as proposals that can be applied, rejected, or edited. Undo/redo further reinforces reversibility.

Responsive behavior intentionally changes the density of the workspace rather than simply shrinking it. At desktop widths the full production layout is visible; at tablet widths navigation and panels become adaptive; below 768px the experience is reduced to a viewer/light-editing model rather than forcing a dense desktop editor onto a small screen.

Trade-off: The professional model has a higher cognitive load than Prototype 1. More controls, panels, and timeline information require more familiarity, but this is exchanged for speed, precision, and direct creative control.

Prototype 3 --- Code Flow Signal

Target: Creative explorer, experimental filmmaker, indie creator, game/concept designer, or anyone beginning with an incomplete idea.

Color: more neon light color and retro + cyberpunk theme (spark creativity instead of plain color)

The core principle is **"exploration over execution."** Instead of asking the user to fill in a conventional video-generation form, Creative Signal begins with a **Signal**: a thought, voice input, text, sketch, or image.

Its core loop is:

Signal → Discover → Combine → Generate → Explore → Branch → Stabilize → Video

The Creative Universe is the central interaction model. Ideas become visual nodes that can be explored, selected, combined, and branched. A synthesis interaction allows the creator to combine two ideas and choose a creative direction such as Unexpected, Cinematic, Dreamlike, Dark, Chaotic, Minimal, or Surreal. Branching preserves alternative directions instead of forcing the creator into a single linear workflow.

The retro-futurist terminal aesthetic supports this mental model: the user feels as if they are entering a creative machine and discovering possibilities rather than operating a conventional editor. However, the visual style is not the product concept by itself; the important innovation is the **combination + branching workflow**.

Voice is deliberately presented as a simulated prototype interaction, with an explicit text fallback. Gesture-based combination also has non-gesture interaction paths so that the experimental interaction model does not become an accessibility barrier.

Trade-off: Creative Signal is intentionally less predictable than the other prototypes. Exploration can produce ambiguity and a larger possibility space, so it is less efficient when the user already knows

exactly what video they want. Its value is helping the creator discover the concept first, then stabilizing it into a storyboard and video.

Why the three prototypes belong together

The prototypes form a progression rather than competing versions of the same interface:

Code Flow: "I have an idea. Help me make a video."

Code Flow Pro: "I know how to make videos. Help me make them faster."

Code Flow Signal: "I do not know exactly what this could become. Help me explore."

Together they demonstrate that an AI video platform can support different levels of expertise and different creative mindsets without forcing every user into the same workflow.